

CODE SPORT



In this month's column, we discuss a few computer science interview questions.

In last month's column, we had discussed the emergence of a new set of deep learning models that support external knowledge injection. We take a break from that discussion this month, and discuss a set of computer science interview questions since a few readers have asked us for help in interview preparation. In today's column, we will cover a few questions that focus on basic computer science and machine learning concepts.

1. Given an arbitrarily long character string, you are asked to find out the most frequently occurring and the least frequently occurring character. Write a C function to solve this problem, given a character string of size N . Now you are told that instead of being a fixed size character string, you are given a stream of characters that are coming dynamically as input. At any point in time, you should be able to print the most frequent and least frequent characters in the stream (note that if two or more characters have the same highest frequency, you can choose to print any of them as the most frequent characters). What is the time and space complexity of your algorithm?
2. You have been given all the stock market transactions made by a user in his/her stock exchange account — buying and selling shares. Each transaction consists of the details on: (a) name of the company, (b) type of transaction (buy/sell), (c) number of shares bought/sold, (d) price of each share bought/sold, and (e) date of the transaction. You also have access to the stock market information in terms of the price of each company's stock at the end of every day. The user would like to know which shares can be sold at a profit of 10 per cent or more on each day. You are asked to write a program, which examines the closing price of the previous day's stock market prices, and provide recommendations to the user on which of the equity holdings can be sold that day for a profit of 10 per cent or more. Your recommendation would contain the company name, number of shares to sell, the sale price, and the percentage of profit that the user will make if he sells at that price. What is the time complexity of your algorithm, assuming that you are given N transactions in total corresponding to M different companies?
3. Consider the above question. Stock markets are quite volatile and the price of company stocks can fluctuate considerably during the day. You are now told that instead of using the previous day's closing price, your application should read in real-time (the current real-time price of the stocks that the user holds), and then recommend which stocks the user can sell at that point of time for a 10 per cent profit. Assume that you have stock market prices available as a real-time feed at every 60 second interval. How would you modify your solution for question (1) to satisfy this criterion?
4. You are given two sentences, S_1 and S_2 , each containing N and M words respectively. You are now asked to convert S_1 into S_2 by either deleting a word or adding a new word. (Modifying word1 to word2 can be considered as word deletion followed by word addition.) Each insert operation has a cost of +2 and each delete operation has a cost of +1. Given two arbitrary sentences S_1 and S_2 , the edit distance of (S_1 , S_2) is the minimum total cost of operations needed to convert S_1 into S_2 . You are asked to write a function to compute the edit distance. What is the time and space complexity of your function?
5. When we input a query into a search engine such as Google, we get an 'auto complete' suggestion for it, based on past queries that have been tried on the search engine. Can you explain how you would design an auto-complete mechanism that will scale well across the millions of queries that can happen concurrently at a time? What is the